

Tutorial_2_Solution.pdf
PDF

Prepare a formal academic lecture scribe intended to function as exam-oriented reference material for a closed-notes examination.

Use only the attached lecture slides/PDF and the corresponding relevant textbook material as the source of information. The scribe must remain strictly faithful to these materials and must not introduce information that is not explicitly contained in the provided context.

The lecture includes conceptual explanations and problem-solving components. Reconstruct the material accurately and completely according to how it appears in the lecture.

The lecture scribe must:

- Follow the same logical order and progression of topics used in the lecture materials.
- Include all definitions, symbols, and mathematical notation introduced in the lecture.
- Clearly identify all assumptions and conditions when they are presented.
- State all theorems, propositions, or formal results exactly as they appear in the lecture.
- Reproduce proofs, derivations, and worked examples step by step whenever they occur in the lecture material.
- Maintain the logical relationships between concepts, results, and example solutions.
- Ensure mathematical precision and clarity throughout the document.
- Use clear section headings and structured organization so that the document can be used effectively for exam revision.

The lecture scribe must not:

- Add interpretations, intuitive explanations, or commentary that are not explicitly stated in the lecture materials.
- Introduce new examples, alternative solution methods, or additional problems.
- Skip intermediate steps in proofs, derivations, or example solutions that are shown in the lecture.
- Modify or reorganize the lecture material in a way that changes its logical sequence.

The final document should accurately reflect what was taught in the lecture so that a student can rely on it as a complete reference when revising for an examination.

Below is a **formal lecture/tutorial scribe** constructed strictly from the uploaded tutorial solution document for **CSE 400: Fundamentals of Probability in Computing**. The material follows the same logical order, notation, and derivations presented in the source.

CSE 400 – Fundamentals of Probability in Computing

Tutorial 2 – Formal Scribe

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Date: February 20, 2026

Source: Tutorial Solution Document  Tutorial_2_Solution

1. Random Point on a Line Segment

Problem

A point is chosen at random on a line segment of length L . Determine the probability that the **ratio of the shorter segment to the longer segment** is less than $\frac{1}{4}$.  Tutorial_2_Solution

Interpretation

Let the point be chosen at a distance x from one end such that

$$0 < x < L$$

This divides the segment into two parts:

- Segment 1: x
- Segment 2: $L - x$

Define the ratio

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We require

$$R < \frac{1}{4}$$

Case Analysis

Case 1: $x \leq \frac{L}{2}$

Shorter segment = x

Longer segment = $L - x$

$$\frac{x}{L - x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter segment = $L - x$

Longer segment = x

$$\frac{L - x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total valid length

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since the point is uniformly distributed over length L ,

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

2. Classification of Image Pixels Using Normal Distributions

Given

White region W :

$$X|W \sim N(4, 4)$$

$$\sigma_1 = 2$$

Black region B :

$$X|B \sim N(6, 9)$$

$$\sigma_2 = 3$$

Prior probabilities

$$P(B) = \alpha$$

$$P(W) = 1 - \alpha$$

Observation:

$$X = 5$$

The probability of error is the same regardless of classification.

Thus,

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Using Bayes' Theorem

$$P(B|X = 5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1 - \alpha) f(5|W)}$$

Normal density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Evaluate densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute into the equation:

$$\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} \bigg/ \left(\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} + \frac{(1 - \alpha) e^{-1/8}}{2\sqrt{2\pi}} \right) = \frac{1}{2}$$

Simplifying:

$$\frac{3\alpha}{3\alpha + 2(1 - \alpha)e^{-5/72}} = \frac{1}{2}$$

Solving:

$$6\alpha = 3\alpha + 2(1 - \alpha)e^{-5/72}$$

$$3\alpha = 2(1 - \alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Thus,

$$\alpha \approx 0.38$$

3. Optimal Location of a Fire Station

(a) Uniform Fire Locations

Let

$$X \sim \text{Uniform}(0, A)$$

PDF

$$f_X(x) = \frac{1}{A}, \quad 0 \leq x \leq A$$

Goal:

$$\min_a E[|X - a|]$$

Expected Value

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

After evaluation:

$$E[|X - a|] = \frac{2a^2 - 2aA + A^2}{2A}$$

Optimization

$$\frac{d}{da} \left(\frac{2a^2 - 2aA + A^2}{2A} \right) = \frac{1}{2A} (4a - 2A)$$

Set derivative to zero:

$$4a - 2A = 0$$

$$a = \frac{A}{2}$$

Thus, the fire station should be located at the **midpoint of the road**.

(b) Exponential Fire Locations

$$X \sim \text{Exponential}(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}$$

Expected value

$$E[|X - a|] = \int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Taking derivative:

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set to zero:

$$1 - 2e^{-\lambda a} = 0$$

$$e^{-\lambda a} = \frac{1}{2}$$

Taking logarithm:

$$-\lambda a = \ln\left(\frac{1}{2}\right) = -\ln 2$$

$$a = \frac{\ln 2}{\lambda}$$

4. Lifetime of Mixed Battery Types

Two battery types exist:

$$T = i, \quad i = 1, 2$$

Lifetimes:

$$X|T = i \sim \text{Exponential}(\lambda_i)$$

Probabilities

$$P(T = i) = p_i$$

Goal:

$$P(X > t + s | X > t)$$

Law of Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Conditional Probability

$$\begin{aligned} P(X > t + s | X > t) &= \frac{P(X > t + s)}{P(X > t)} \\ &= \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}} \end{aligned}$$

5. Conditional Distribution of Poisson Variables

Let

$$X \sim \text{Poisson}(\lambda_1)$$

$$Y \sim \text{Poisson}(\lambda_2)$$

independent.

Goal:

$$P(X = k | X + Y = n)$$

Conditional Probability

$$P(X = k | X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Because of independence:

$$P(X = k, Y = n - k) = P(X = k)P(Y = n - k)$$

Substituting Poisson PMFs:

$$= \frac{\left(e^{-\lambda_1} \frac{\lambda_1^k}{k!} \right) \left(e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!} \right)}{e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}}$$

Simplifying:

$$P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

This is a **Binomial distribution**.

6. Transformation of a Uniform Random Variable

Given

$$X \sim \text{Uniform}[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

Transformation:

$$g(x) = \begin{cases} x & -2 \leq x \leq -1 \\ 0 & -1 < x < 1 \\ x & 1 \leq x \leq 2 \end{cases}$$

Let

$$Y = g(X)$$

CDF $F_Y(y)$

Piecewise evaluation:

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \leq y < -1 \\ \frac{1}{4} & -1 \leq y < 0 \\ \frac{3}{4} & 0 \leq y < 1 \\ \frac{y+2}{4} & 1 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF $f_Y(y)$

For one-to-one regions:

$$f_Y(y) = f_X(y) \left| \frac{dx}{dy} \right|$$

Since $dx/dy = 1$:

$$f_Y(y) = \frac{1}{4}$$

for

$$-2 < y < -1$$

and

$$1 < y < 2$$

with

$$P(Y = 0) = \frac{1}{2}$$

7. Stock Price Model

Price evolution:

$$S_{1000} = S_0 u^k d^{1000-k}$$

where

- $u = 1.012$
- $d = 0.990$
- $p = 0.52$

Condition

$$S_{1000} \geq 1.30 S_0$$

$$u^k d^{1000-k} \geq 1.30$$

Taking logarithms:

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

Solving:

$$k \geq 470$$

Since

$$k \sim \text{Binomial}(1000, 0.52)$$

Probability:

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

8. Exponential Repair Time

$$X \sim \text{Exponential}(\lambda = \frac{1}{2})$$

PDF

$$f(x) = \frac{1}{2}e^{-x/2}, \quad x \geq 0$$

(a) Repair Time Exceeding 2 Hours

$$\begin{aligned} P(X > 2) &= \int_2^{\infty} \frac{1}{2}e^{-x/2} dx \\ &= e^{-1} \\ &\approx 0.3679 \end{aligned}$$

(b) Conditional Probability

$$\begin{aligned} P(X \geq 10 | X > 9) &= \frac{P(X \geq 10)}{P(X > 9)} \\ P(X \geq 10) &= e^{-5} \\ P(X > 9) &= e^{-9/2} \\ P(X \geq 10 | X > 9) &= e^{-1/2} \end{aligned}$$

9. Distribution of $Z = \sqrt{X^2 + Y^2}$

Let X and Y be independent.

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Define

$$Z = \sqrt{X^2 + Y^2}$$

CDF

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(X^2 + Y^2 \leq z^2) \end{aligned}$$

Region:

$$x^2 + y^2 \leq z^2$$

Thus,

$$F_Z(z) = \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule:

$$\frac{d}{dz} \sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Thus

$$f_Z(z) = z \int_{-z}^z f_X(x) \frac{f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})}{\sqrt{z^2 - x^2}} dx$$

for

$$z > 0$$

10. Joint Distribution of M_n and M_{n+1}

Let

$$X_1, X_2, \dots$$

be i.i.d. with CDF F .

Define

$$M_n = \max(X_1, \dots, X_n)$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

Distribution of M_n

$$P(M_n \leq x) = P(X_1 \leq x, \dots, X_n \leq x) = F(x)^n$$

Joint CDF

$$P(M_n \leq x, M_{n+1} \leq y)$$

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_{n+1} \leq y) = F(y)^{n+1}$$

Final Joint CDF

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y) & x \leq y \\ F(y)^{n+1} & x > y \end{cases}$$

If you want, Datt, I can also **convert this into a clean LaTeX document (exactly like the assignment format your professor expects)** so you can directly compile and submit it.



Sources

